A SCREENING TOOL FOR A PAY-EQUITY REGULATOR

Pay-Equity Audit with Supervised Learning

−5.4 pts

Given the same job, education and hours, women earn over $50K, 5.4 percent less often than expected.

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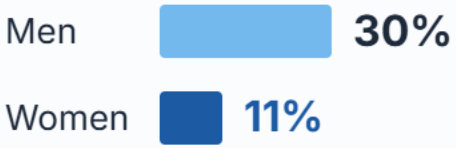
Supervised Learning & Model Engineering · Prof. Ilana Sebag · Albert School · October 2026

Our role: a pay-equity regulator

Business question: which employers should we investigate, and on what evidence?

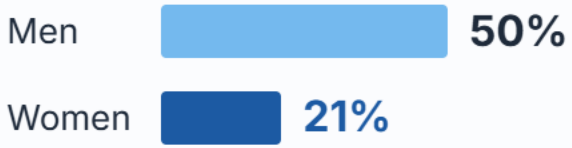
All workers

Share earning more than $50K



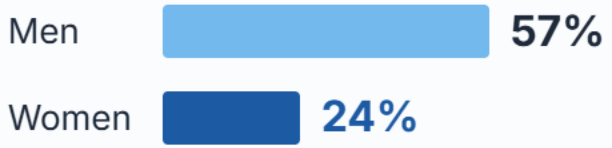
Same degree: Bachelors

Share earning more than $50K



Same job: managers

Share earning more than $50K



So what? The gap does not disappear when we compare people with the same degree or the same job. Raw gaps are not enough: under “equal pay for equal work”, we must compare each person with what their job-related profile predicts.

Adult census data, full dataset (48,842 people). Men: light blue · Women: navy.

What the data tells us

48,842

people, 1994 US census

Enough data to train, tune and still keep 9,769 people aside for a final test.

→ **Stratified 60 / 20 / 20 split; the test set is opened once, at the end.**

24%

earn more than $50K

A model that always answers “no” is 76% accurate, and useless.

→ **We judge models on average precision and recall, not accuracy.**

6%

occupation unknown

These people earn more than $50K far less often: 9% against 25%.

→ **“Missing” is kept as its own category instead of being filled in.**

So what? The data is rich but imbalanced and imperfect. Each of these facts shaped a decision, so that the numbers we report are honest.

What the model may use

USED · JOB-RELATED FACTORS

- Age
- Education (years)
- Occupation
- Hours per week
- Workclass
- Investment income

EXCLUDED · PROTECTED OR PROXIES

- ~~Sex~~
- ~~Race~~
- ~~Native country~~
- ~~Relationship~~
- ~~Marital status~~

> "Husband" = 99.99% men: relationship encodes sex.

ADDED · 5 NEW FEATURES

Works overtime (> 40 h): 40% earn > $50K vs 17%

Part-time (< 35 h): 7% vs 27%; 28% of women vs 12% of men

Has investments, net and log capital gain: 58% vs 19%

So what? The exclusions happen inside the pipeline, so sex and race can never reach the model, even in production. Part-time work is a legitimate reason to earn less, and the model now accounts for it.

Also dropped: fnlwgt (census sampling weight) and education (exact duplicate of education years).

How the audit works

Employee profile
age, education, occupation, hours, workclass, investments

→

Reference pay model
trained on the population, never sees sex or race

→

Expected probability
of earning more than $50K for this profile

For each group: **actual rate** − **expected rate** = **unexplained gap**

1 · No protected attributes

otherwise the model learns the discrimination and hides it

2 · Good ranking

the expected rate must follow real differences in qualifications

3 · Calibrated probabilities

the expected rate is an average of probabilities, so they must be right

A negative gap means the group earns less than its qualifications predict: a reason to investigate, not a verdict.

Three models, one winner

Average precision on the validation set, after tuning (main metric: focuses on the rare > $50K class; random = 0.24)

Logistic Regression

0.692

Simple and transparent, but one linear score misses interactions.

Random Forest

0.752

Biggest jump: the value of a degree depends on the job and on age.

LightGBM · selected

0.776

Best ranking and recall, trains in about 1 second.

Imbalance: class weights

Same gain as undersampling or SMOTE, but keeps every real person and creates no synthetic ones.

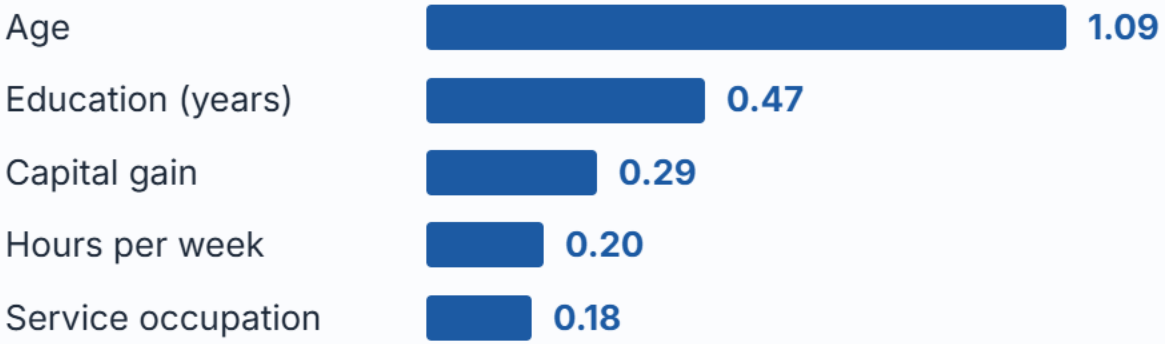
Tuning: Optuna + 5-fold CV

Only +0.005 AP: the choice of model family mattered far more than its settings.

One scikit-learn Pipeline: drop excluded columns, add 5 features, impute, scale, one-hot, model · every run logged in MLflow

What drives the predictions

Mean |SHAP value|: average impact on a prediction



ONE PREDICTION EXPLAINED

Age 50, professional degree, self-employed. Predicted **91%** chance of earning more than $50K, but earns less.

It looks like underpayment, yet there is no employer to blame.

→ **Flags are leads for inspectors, not verdicts.**

So what? The model prices a profile the way a pay grid would: experience, qualification, investment income, working time and job. These are factors an employer cannot dispute.

SHAP (TreeExplainer) on 2,000 validation people. SHAP describes what the model uses, not what causes income.

Can we trust the probabilities, and act on them?

1 · FIX THE PROBABILITIES

Average predicted chance of earning > $50K



Class weights inflated every probability. Left as is, **every group would look about 13 points underpaid.** Calibration error: 0.135 → 0.009.

2 · DECIDE WHEN TO FLAG

Flag an employee when the calibrated chance is ≥ 50% (“more likely than not”) but they earn less

82%

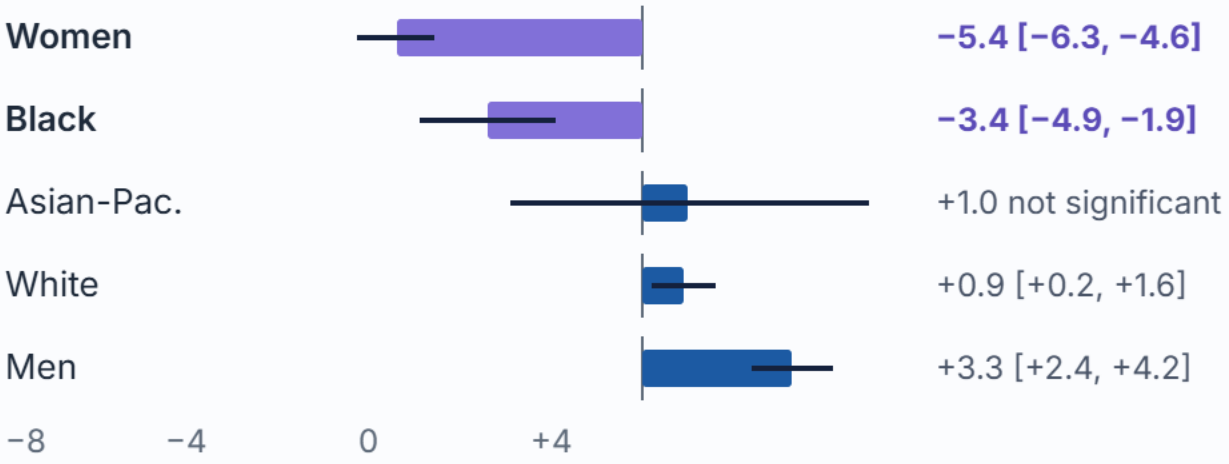
When the model expects > $50K, it is right 82% of the time. The F1-optimal threshold (0.39) would drop this to 68%: more false alarms for each inspector.

So what? Calibration makes the comparison between groups fair. A strict threshold keeps inspectors on strong cases, because their time is the scarce resource.

Isotonic calibration learned on out-of-fold training predictions · threshold chosen on validation · test set untouched until the end

The unexplained gap

Actual minus expected share earning > $50K, in points (bar), with its 95% range (line)



WHAT IT MEANS

Women: 1 in 3 whose profile predicts more than $50K does not get it. For men: 1 in 7.

Black employees: the gap is spread over moderate profiles, so it shows for the group more than in individual flags.

Both ranges stay clearly below zero.

So what? Two significant gaps, for women and for Black employees: grounds to open investigations. Unexplained is not proven, so the employer gets the chance to justify the gap.

Untouched test set (9,769 people) · 95% bootstrap confidence intervals · groups under 100 people not interpreted

Why not use all columns?

Before: all columns

including sex, race, relationship, marital status

ROC-AUC	Gap, women	Gap, Black
0.930	+0.2	+1.1

Every employer would look fair.

After: job-related only (ours)

no sex, no race, no proxies

ROC-AUC	Gap, women	Gap, Black
0.893	−5.4	−3.4

The gaps become visible.

So what? The full model is more accurate because it has learned the discrimination, and so it hides it. Our mitigation costs 0.04 ROC-AUC, but it is what makes the audit possible.

Also rejected: post-processing (ThresholdOptimizer). Equalising error rates between groups would erase the very signal we measure, and would need sex and race at prediction time.

Same tuned LightGBM, calibration and threshold; only the input columns differ · untouched test set

How solid is the result?

CHECK 1 · NEW DATA

0.893

ROC-AUC, validation and test

Same performance and calibration (error 0.009) on people never seen before.

→ **Our choices did not overfit.**

CHECK 2 · LEAKY COLUMNS

−6.6

gap for women without capital income

Capital gains are part of total income. Without them: women −6.6, Black employees −3.8.

→ **The conclusion holds; the gaps even grow.**

CHECK 3 · HIDDEN PROXIES

0.79

ROC-AUC to guess sex from our inputs

Occupation and hours still carry sex. Unequal access to good jobs is "explained away".

→ **Our gaps are a lower bound.**

So what? The evidence is conservative: if anything, the audit under-states the problem. That is the right side to err on when the result can lead to an investigation.

All checks on the untouched test set · gaps in points, 95% bootstrap confidence intervals

From model to tool, and our recommendation

FASTAPI SERVICE

POST /audit

An inspector sends a company's employee records and gets the gap per group, its 95% range and the flagged employees.

```
women: gap −6.0, range [−7.8, −4.0]
flagged as potentially underpaid: 52
```

Simulated company of 2,000 employees

POST /predict

Same profile, any sex or race: same answer (tested: 65.2% for both).

RECOMMENDATION

Screen every company's records. Open an investigation when a group's gap is significantly negative, and use the flags to choose which files to review.

LIMITS

Unexplained is not proven: seniority, region and performance are not in the data.
1994 census of individuals, not companies.
Total income above $50K, not exact salary.

Thank you! · Code: github.com/rcacarie-png/Adult-Income-Prediction